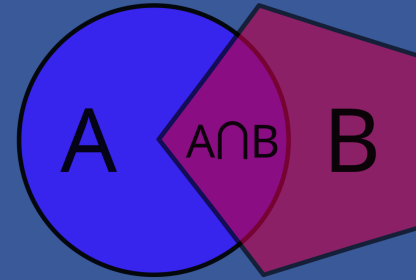
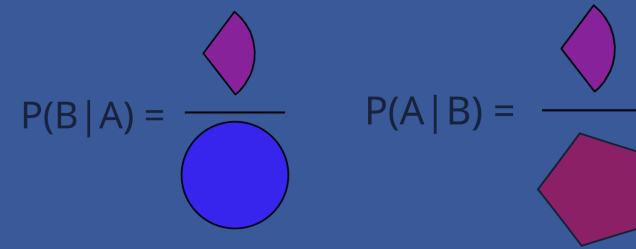
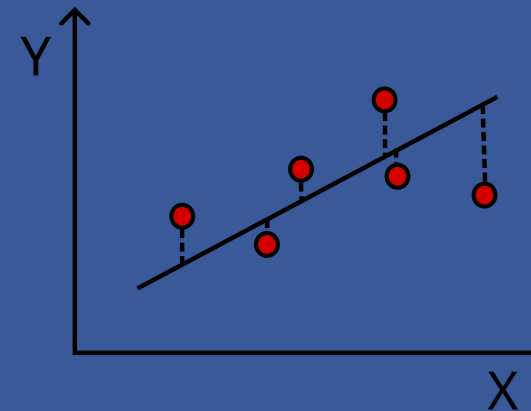


Maschinelles Lernen für die Chemie - Exercise 2



$$P(B|A) = \frac{\text{Area of } A \cap B}{\text{Area of } A}$$
$$P(A|B) = \frac{\text{Area of } A \cap B}{\text{Area of } B}$$


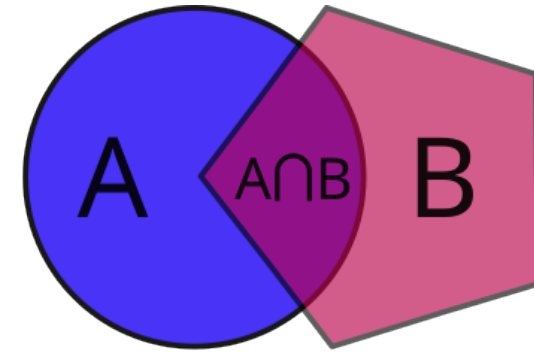


Bayes' Theorem, Data Types and Regression

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Probabilities and Notation

- $P(A)$: Probability of observing event A
- $P(B)$: Probability of observing event B
- $P(A|B)$: **Conditional** probability, the probability of event A occurring given that B is true



$$P(B | A) = \frac{\text{Area of } A \cap B}{\text{Area of } A}$$
$$P(A | B) = \frac{\text{Area of } A \cap B}{\text{Area of } B}$$

The diagram shows the conditional probability formulas for P(B | A) and P(A | B). For P(B | A), the numerator is represented by a small pink shape (the intersection) and the denominator by a blue circle (set A). For P(A | B), the numerator is represented by a small pink shape (the intersection) and the denominator by a pink pentagon (set B).

Bayes' Theorem

Mathematical rule for **inverting conditional probabilities**

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

Proof:

$$P(A) \cdot P(B|A) = \frac{\text{Blue Circle}}{100\%} \cdot \frac{\text{Pink Diamond}}{\text{Blue Circle} + \text{Pink Diamond}} = \frac{\text{Pink Diamond}}{100\%}$$

$$P(B) \cdot P(A|B) = \frac{\text{Pink Diamond}}{100\%} \cdot \frac{\text{Blue Circle}}{\text{Blue Circle} + \text{Pink Diamond}} = \frac{\text{Blue Circle}}{100\%}$$

$$= P(A) \cdot P(B|A)$$

$$P(A|B) = \frac{P(A) \cdot P(B|A)}{P(B)}$$

Bayes' Theorem – Example

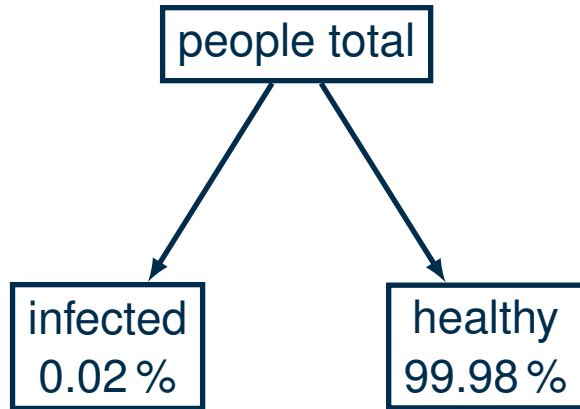
Testing for an infection (T : test positive, K infected people, G healthy people)

- Prevalence (people who get infected): $P(K) = 0.0002$
- Sensitivity (positive test if person is infected): $P(T|K) = 0.95$
- Specificity (positive test if person is healthy): $P(T|G) = 0.01$
- Amount of tested people: 100 000
- What is the probability that a random positively tested person is actually infected?

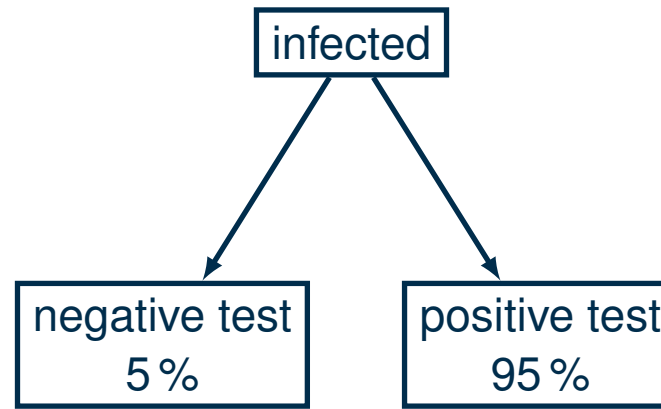
Bayes' Theorem – Example

Testing for an infection (T : test positive, K infected people, G healthy people)

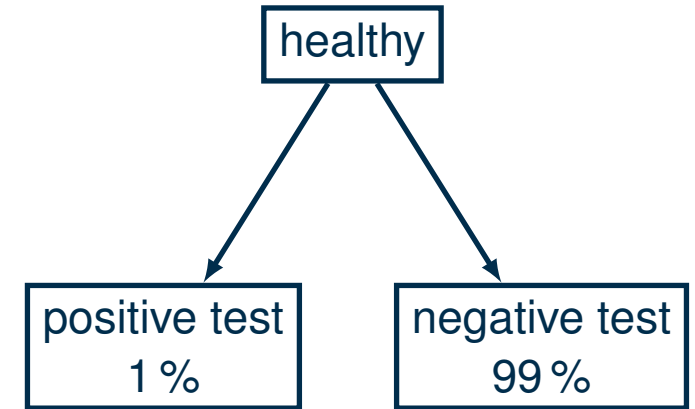
Prevalence $P(K) = 0.0002$



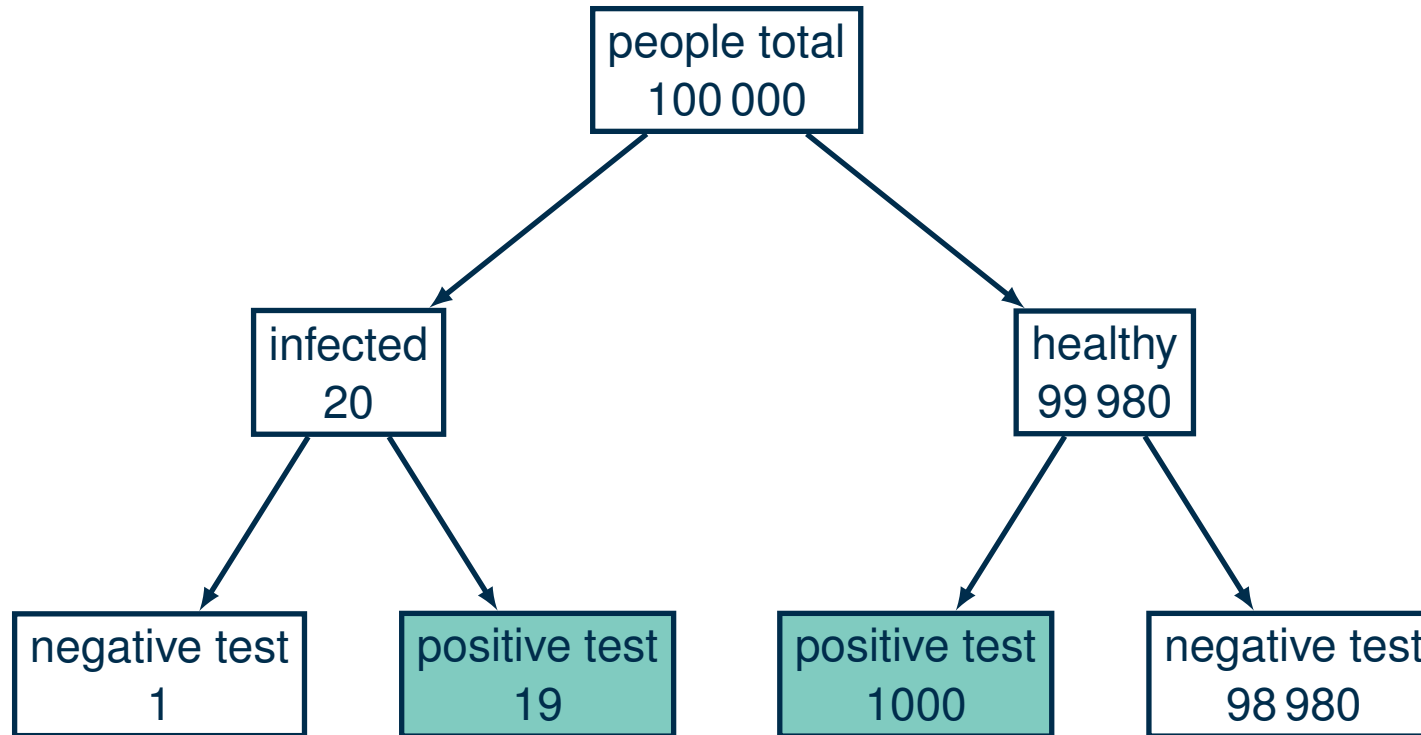
Sensitivity $P(T|K) = 0.95$



Specificity $P(T|G) = 0.01$



Bayes' Theorem – Example



$$P(K|T) = \frac{P(T|K) \cdot P(K)}{P(T)}$$

Bayes' Theorem – Example

Solution:

$$P(K|T) = \frac{P(T|K) \cdot P(K)}{P(T)} = \frac{0.95 \cdot 0.0002}{0.95 \cdot 0.0002 + 0.01 \cdot 0.9998} \approx 0.0186$$

with the probability for the positively tested people

$$P(T) = P(T|K) \cdot P(K) + P(T|G) \cdot P(G)$$

What Are Data Types?

- Each variable has a data type
- Other programs: data type has to be set by the user
Python: **dynamic typing**, data type is declared automatically
- Data type is important, because it determines:
 1. the amount of memory used for this variable
 2. the attributes of the variable (i. e. elements of a list can be accessed, accessing the elements of an integer doesn't make sense)

Examples for Data Types in Python

- Integer: `x = 2`
- Floats: `pi = 3.1415`
- String: `"Some random text"`
- List: `my_list = [1, 2, 3]`
- Dictionary: `my_dict = {key: value}`
- Boolean: `condition = True`

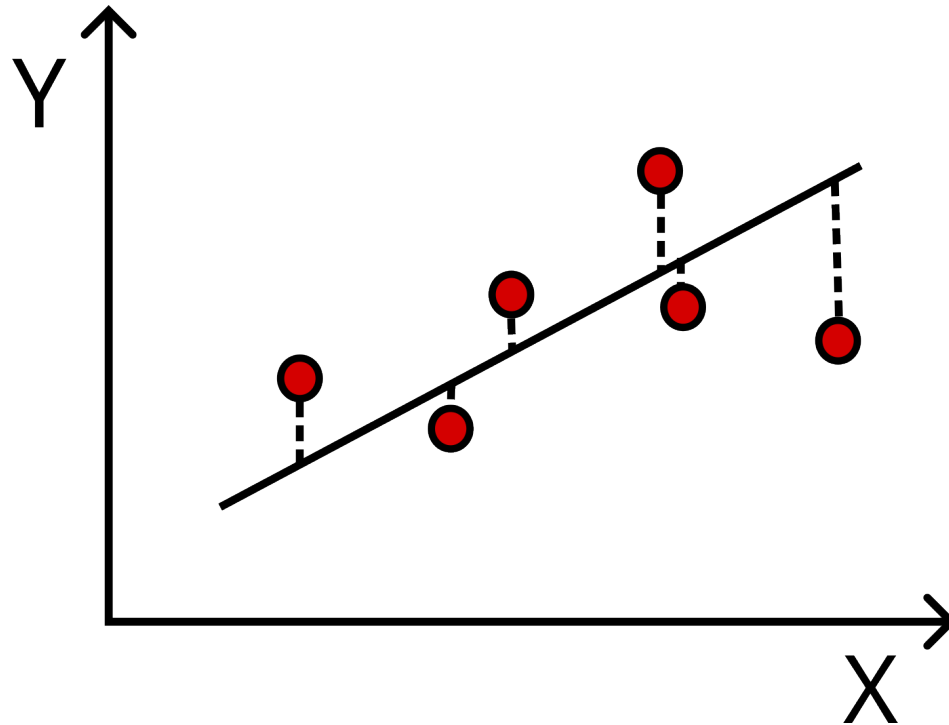
Type Hints

- Example: `x: int = 2`
- States directly, that `x` should be an integer
- In the notebooks: used for hints what to do

```
x: int
y: np.ndarray
# Your code:
x = 100 # Sample size
y = np.arange(100) # Numpy array with all values from 0 to 99
```

Linear Regression

- Input vector (features) X
- Response (label, target) Y



$$f(X) = \beta_0 + \sum_j X_j \beta_j \quad (1)$$

With the set of training data (x_i, y_i) , we minimize the **residual sum of squares**:

$$\text{RSS}(\beta) = \sum_i (y_i - f(x_i))^2 \quad (2)$$

$$= \sum_i \left(y_i - \beta_0 - \sum_j x_{ij} \beta_j \right)^2 \quad (3)$$

Linear Regression

Unique solution for the parameters:

$$\hat{\beta} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$$

- $\mathbf{X}$: $N \times (p + 1)$ matrix, N input vectors as rows, p length of input vectors, 1 in the first position (to account for β_0)
- $\mathbf{y}$: N -vector of outputs in the training set

Exercise:

- Features X : medical information (ten columns)
- Target Y : diabetes or not
- $\mathbf{X}$: $(N_{\text{test}} \times 11)$, first column consisting of ones

Linear Regression – Exercise

- Split the data in two random subsets for training and testing
⇒ Scikit-learn: `train_test_split(feature_array, target_array, test_size=?, random_state=0)`
- Insert column of ones to the training set X
⇒ Numpy array of ones: `np.ones((train_set.shape[0], 1))`
⇒ Prepend the array to the training set: `np.concatenate((ones_array, train_set), axis=1)`
- Calculate $\hat{\beta}$ with the prepended training matrix and the targets:
Matrix multiplication: ⇒ Numpy `np.matmul` or `@`
Invert matrix: ⇒ `np.linalg.inv`
Transpose matrix: ⇒ `matrix.T`

Linear Regression – Exercise

Or completely with Scikit-learn:

- Create a linear regression instance:

⇒ `model: LinearRegression = LinearRegression()`

- Train your model:

⇒ `model.fit(train_set, target_set)`

- Test the model by predicting the targets with the test set:

⇒ `model.predict(test_set)`

- Save the parameters $\hat{\beta}$ as one Numpy array by inserting the interception at position 0 to the other coefficients:

⇒ `np.insert(model.coef_, 0, model.intercept_, axis=0)`

Today's Notebooks

2.1_Numpy_Crashcourse.ipynb

- Optional, from Ex. 1

1.1_LogisticRegressionTutorial.ipynb

- Logistic regression tutorial to predict the subspecies of an iris flower
- Look here, if you need hints for the exercise notebook

1.2_RegressionExercise.ipynb

- Linear regression exercise as explained earlier
- Quantification of the results: refer to tutorial
- Polynomic regression: similar to linear regression, use appropriate Scikit-learn object
- Lasso regression as regularization example, again, different Scikit-learn object but otherwise similar

Bayes' Theorem
○○○○○

Data Types
○○○

Regression
○○○○●

Literature

- Linear Regression [1]
- Bayes' theorem and probabilities [2, 3]

- [1] Trevor Hastie, Tibshirani Robert, and Jerome Friedman. The Elements of Statistical Learning. 2nd ed. Springer Series in Statistics. NY: Springer New York, 2009. ISBN: 978-0-387-84858-7. DOI: <https://doi.org/10.1007/978-0-387-84858-7>.
- [2] Wikipedia contributors. Bayes' theorem. 2025. URL: https://en.wikipedia.org/wiki/Bayes%27_theorem (visited on 05/04/2025).
- [3] Wikipedia contributors. Satz von Bayes. 2025. URL: https://de.wikipedia.org/wiki/Satz_von_Bayes (visited on 05/04/2025).

References